# OMNI CAMERA ROVER | REV E PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

Battery protection

File: battery_protection.kicad_sch

5. Interlock

File: interlock.kicad_sch

4. Regulator

File: regulator.kicad_sch

Flex camera

File: camera.kicad_sch

Cell undervoltage load shutdown

File: cell_undervoltage.kicad_sch

Cell temperature charging window

File: charge_temperature.kicad_sch

Independent discharge cutoff

File: discharge_cutoff.kicad_sch

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

PWR_FLAG




PWR_FLAG




PWR_FLAG




PWR_FLAG




PWR_FLAG




PWR_FLAG




PWR_FLAG




**Nathaniel Sanford**

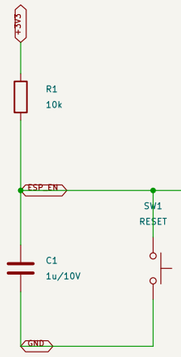
Sheet: /
File: omni_ugv.kicad_sch

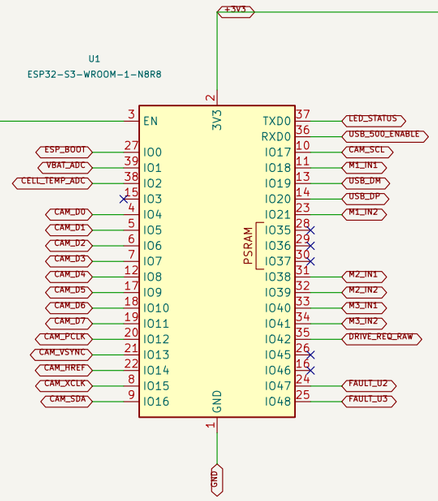
***Title: Omni UGV / Rev E***

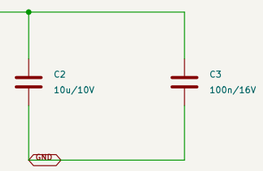
| | | |
|---|---|---|
| Size: A3 | Date: 2026-11-07 | **Rev: E – prototype** |
| KiCad E.D.A. 10.0.8 | | Id: 1/11 |

# Controller, camera and service points

## RESET / enable RC




## Waveshare OV2640 | documented 2 x 9 interface

## Status indicator




## BOOT strap




## Camera reset RC

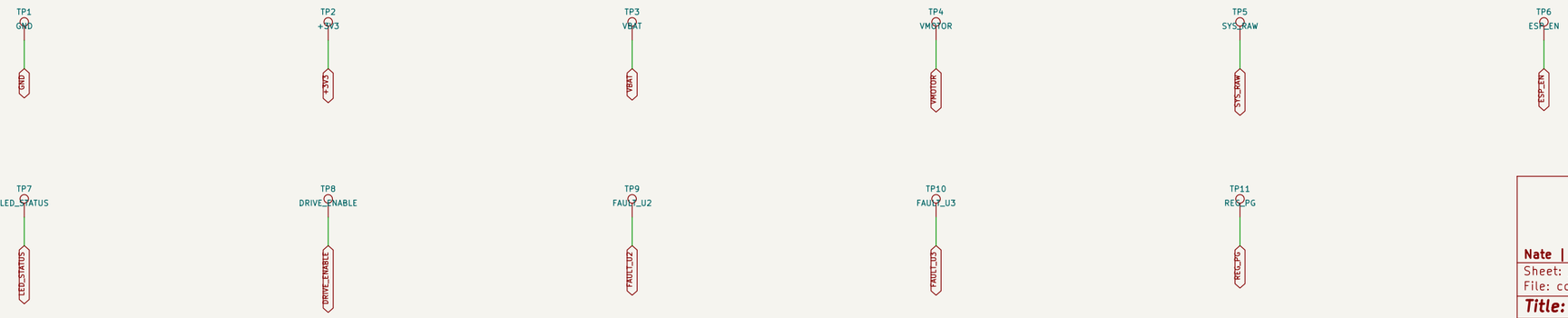






[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

## Service points (named nets continue on other sheets)




**Nate | Omni camera rover**

Sheet: /1. Controller/
File: controller.kicad_sch

***Title: Controller, camera and service points***

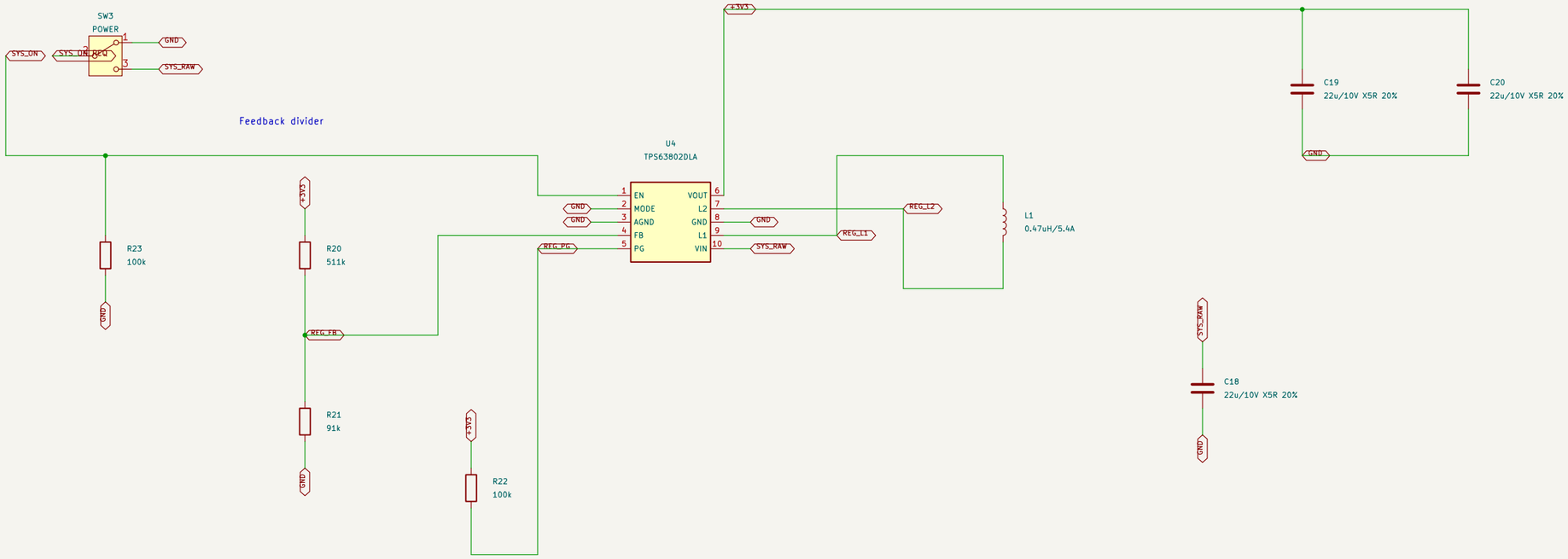
| Size: A3 | Date: | **Rev: E – prototype** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 2/11 |

# 3.3 V buck–boost supply

System enable

Feedback divider

Output capacitors

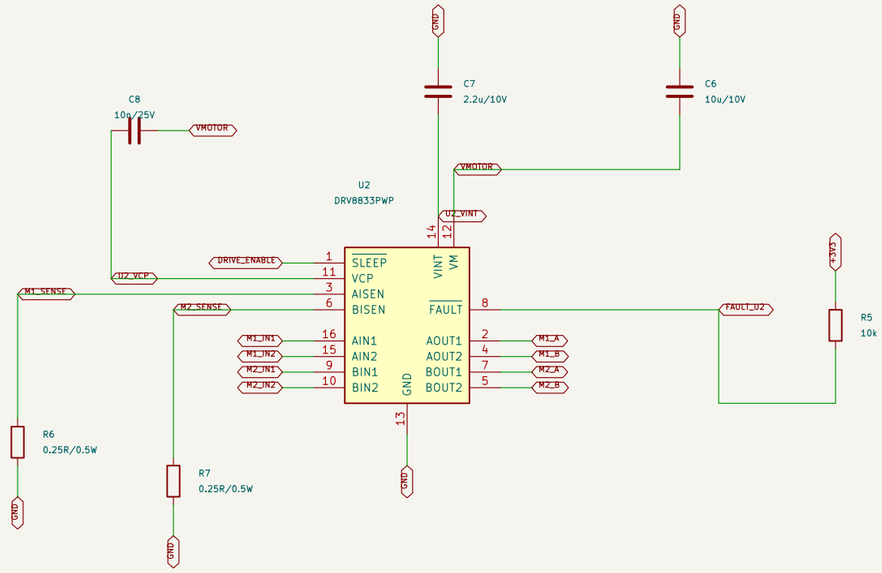



[TARGET] Keep switching loops short. Place input/output capacitors at U4.
[TBD–MEASURE] Verify startup, load transients and capacitor effective capacitance.

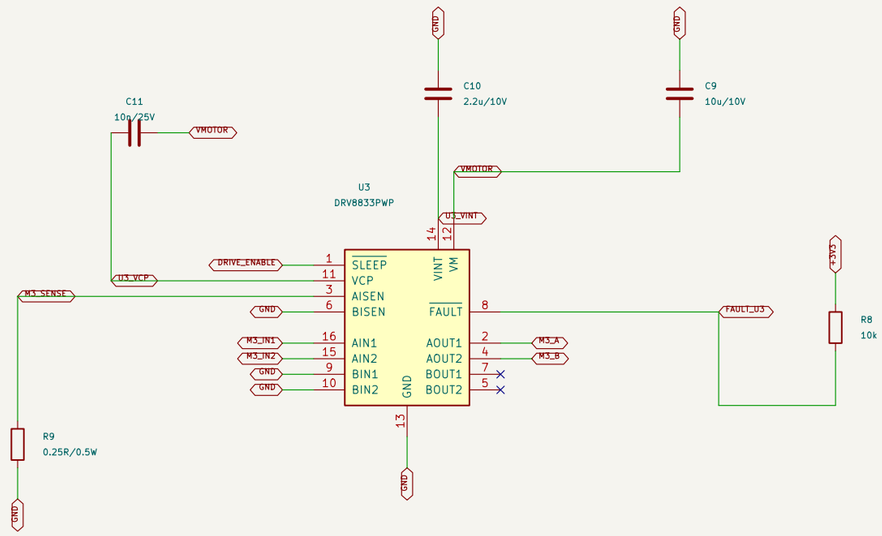
| | | |
|---|---|---|
| **Nate \| Omni camera rover** | | |
| Sheet: /4. Regulator/ | | |
| File: regulator.kicad_sch | | |
| ***Title: 3.3 V buck–boost supply*** | | |
| Size: A3 | Date: | **Rev: E – prototype** |
| KiCad E.D.A. 10.0.8 | | Id: 3/11 |

# Three independent motor channels

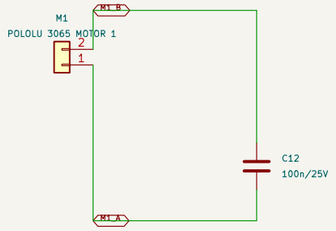
## U2: wheels 1 and 2

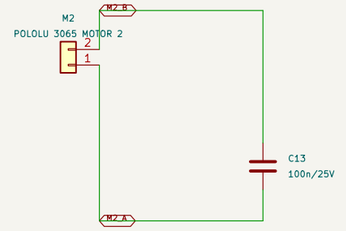


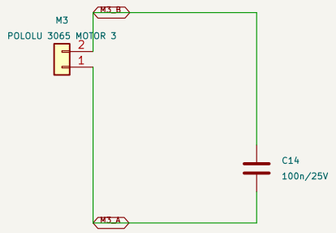

## U3: wheel 3; unused bridge pins retained as designed

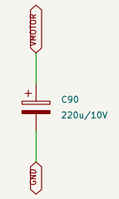



## Motor terminal pads and suppression capacitors



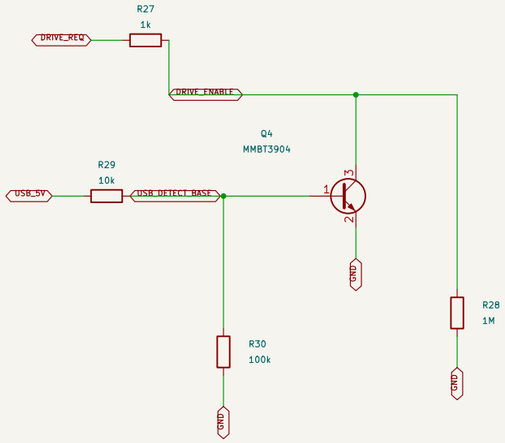




[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

| | |
|---|---|
| Nate \| Omni camera rover | |
| Sheet: /2. Motors/ | |
| File: motors.kicad_sch | |
| **Title: Three independent motor channels** | |
| Size: A3 | Date: |
| KiCad E.D.A. 10.0.8 | **Rev: E — prototype** |
| | Id: 4/11 |

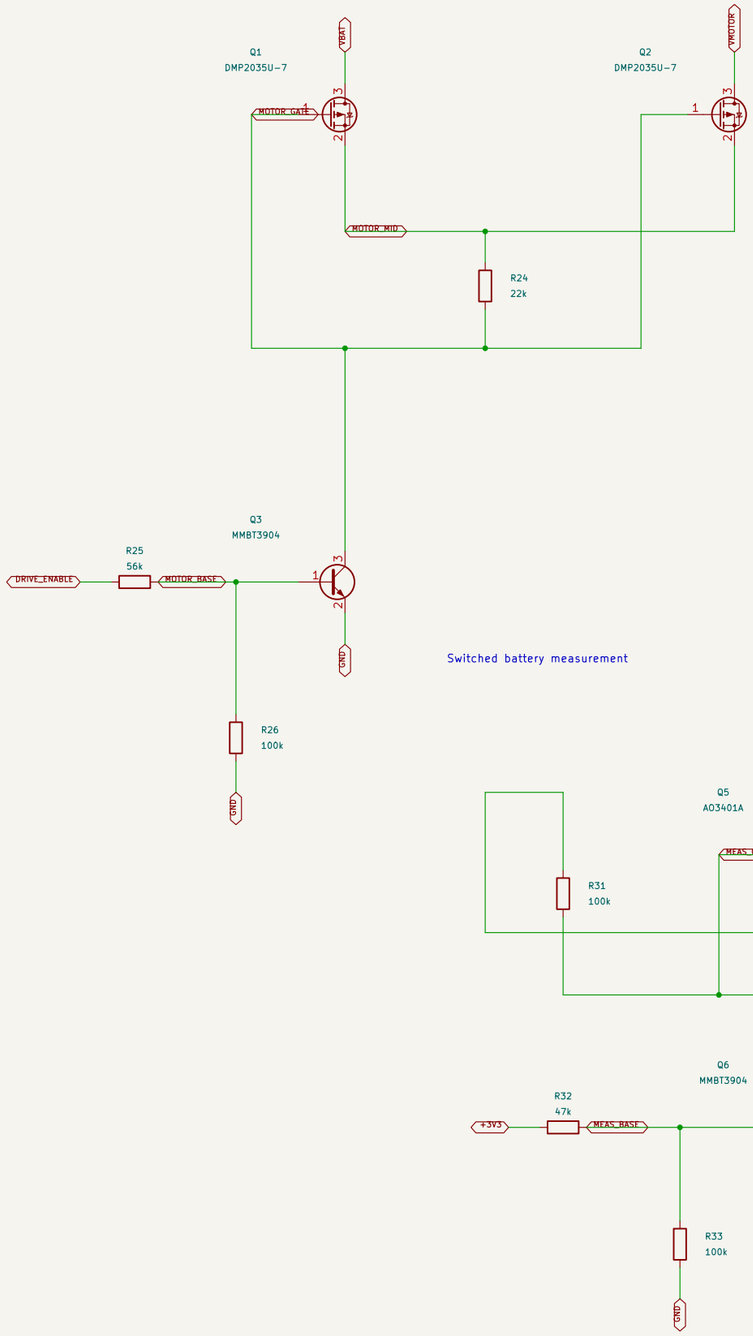
# USB motor interlock and battery measurement

## USB presence clamps DRIVE_ENABLE

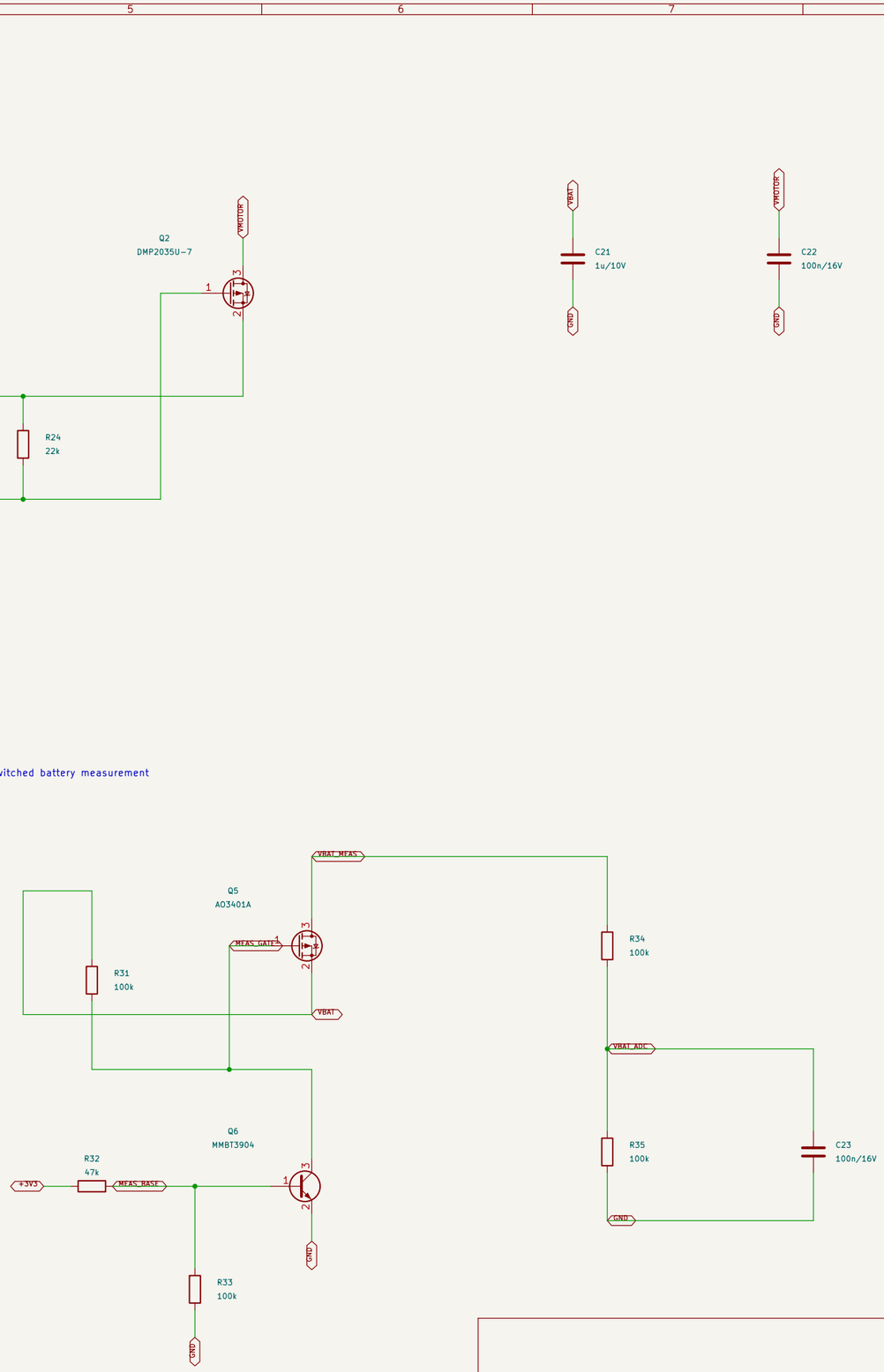



[TARGET] Reset / absent commands keep motors disabled.
[TBD-MEASURE] Verify USB insertion/removal and motor braking transients.

## Back-to-back motor supply switch




## Switched battery measurement







| | |
|---|---|
| Nate \| Omni camera rover | |
| Sheet: /5. Interlock/ | |
| File: interlock.kicad_sch | |
| **Title: USB motor interlock and battery measurement** | |
| Size: A3 \| Date: | **Rev: E – prototype** |
| KiCad E.D.A. 10.0.8 | Id: 5/11 |

# USB input, charging and cell connections

## USB data to ESP32 native USB

J1
USB-C

R10
5.1k

R11
5.1k

U6
USBLC6-2SC6

R12
22R

R13
22R

## Raw cell -> protection sheet

J2
RAW 1S CELL 4.2V MAX

## Charge / system power path

U5
BQ24074RGT

C15
1u/10V

C17
10u/10V

C16
10u/10V

R16
46.4k

R17
100k

R14
3.57k

R15
3.24k

R19
100k

R18
2.2k

D2
CHARGE AMBER

## Cell-bonded charger probe

J4
CELL NTC - CHARGER

[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two-wire 103AT-2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

| | |
|---|---|
| **Nate \| Omni camera rover** | |
| Sheet: /3. Charging/ | |
| File: charging.kicad_sch | |
| ***Title: USB input, charging and cell connections*** | |
| Size: A3 | Date: |
| KiCad E.D.A. 10.0.8 | **Rev: E - prototype** |
| | Id: 6/11 |

# Four-terminal cell protection and cell probes

Kelvin sense: no motor current in S_CELL_N / S_GND traces

R37
100R 1%

C24
100n/16V X7R

R38
20R 1%

U7
BQ298217RUGR

R39
1k 1%

R41
10M

Q7
CSD17308Q3

Q8
CSD17308Q3

R40
10M

TH2 -> J5 -> GPIO2 / ADC1

R42
10k 1%

J5
CELL NTC - DRIVE

R43
1k 1%

TH2
103AT-2 CELL-MOUNTED

C25
100n/16V X7R

R44
10m 1% 1W KELVIN

TH1 -> J4 analog charging window

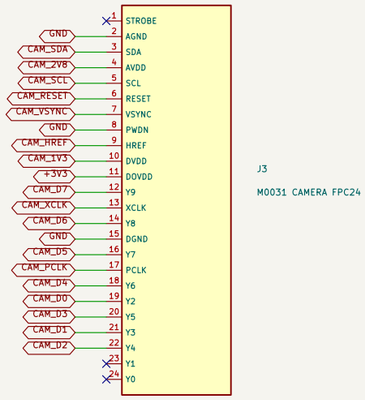
TH1
103AT-2 CELL-MOUNTED

#FLG21
PROTECTION_PWR_FLAG

#FLG20
PROTECTION_PWR_FLAG

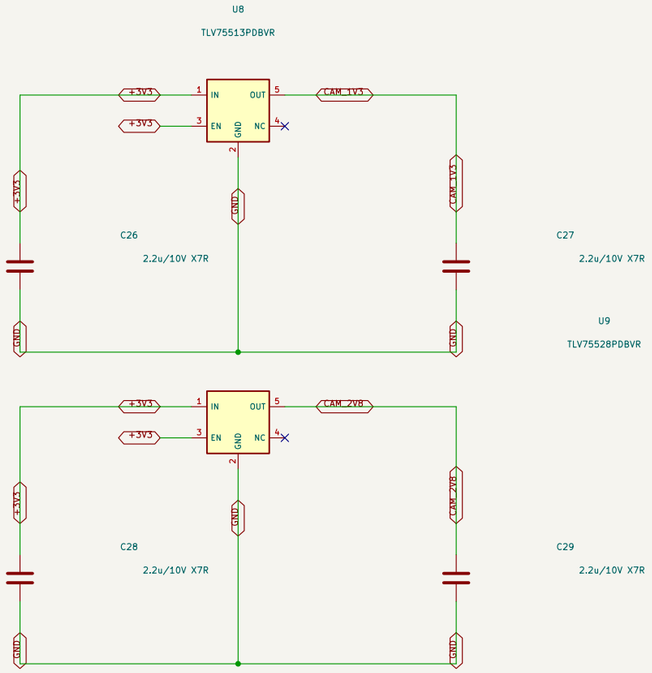
WSK2512 current pads1/2 replace WSL2512. Sense3 -> U7 VSS + C24 return; sense4 -> U7 CS.
The voltage contacts are internal Kelvin pickups, not ground-plane taps.
TH1 and TH2 remain OFF-BOARD, insulated and bonded to the actual cell.

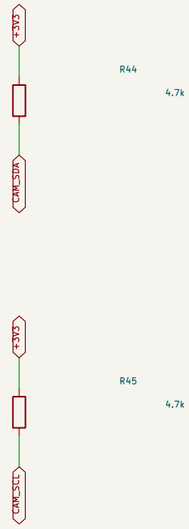
| | |
|---|---|
| **Nate \| Omni camera rover** | |
| Sheet: /Battery protection/ | |
| File: battery_protection.kicad_sch | |
| ***Title: Four-terminal cell protection and cell probes*** | |
| Size: A3 | Date: |
| KiCad E.D.A. 10.0.8 | **Rev: E – prototype** |
| | Id: 7/11 |




J3

M0031 CAMERA FPC24



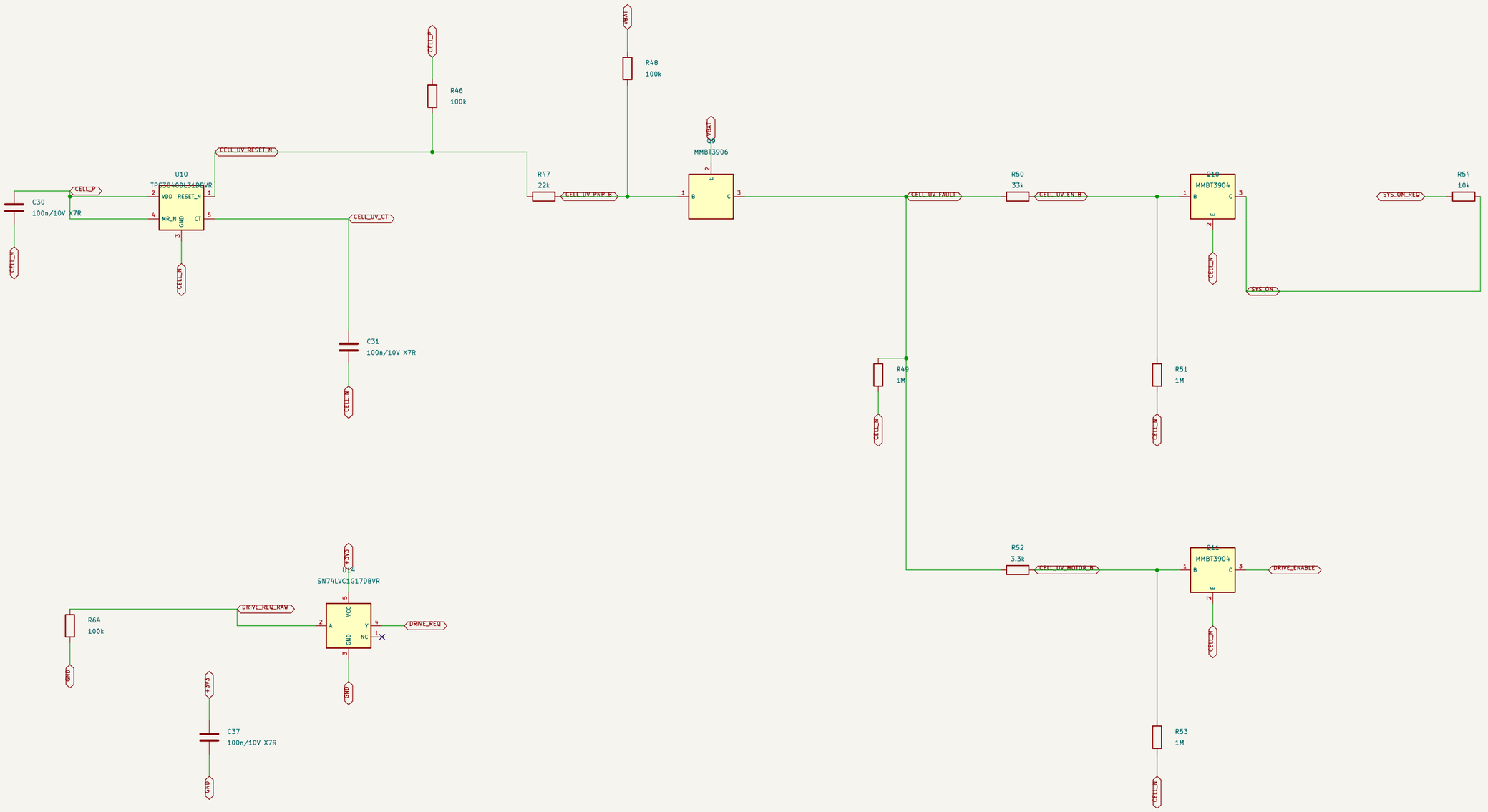




M0031 only. Verify flex contact side, thickness and pin-1 before assembly.
1.3 V core / 2.8 V analog / 3.3 V I-O. No interchangeable generic OV2640 pinout.

| | |
|---|---|
| Sheet: /Flex camera/ | |
| File: camera.kicad_sch | |
| **Title: OV2640 M0031 flex interface** | |
| Size: A3 | Date: |
| KiCad E.D.A. 10.0.8 | **Rev: E – prototype** |
| | Id: 8/11 |

# Independent cell undervoltage shutdown

3.1 V supervisor: load inhibits plus independent discharge cutoff on sheet 11




R46 now pulls RESET_N to raw CELL_P. Q12/Q13 remain effective after protected VBAT disappears.
Q9/Q10/Q11 retain the existing logic and motor inhibits; R64 defaults the motor request low.
U7 retains independent overvoltage/current/short protection and its factory 2.6 V UV threshold.
This is not a zero-leakage storage disconnect: the supervisor, protector and cutoff bias still use cell current.

**Nate | Omni camera rover**

Sheet: /Cell undervoltage load shutdown/
File: cell_undervoltage.kicad_sch

***Title: Independent cell undervoltage shutdown***

| Size: A3 | Date: | **Rev: E – prototype** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 9/11 |

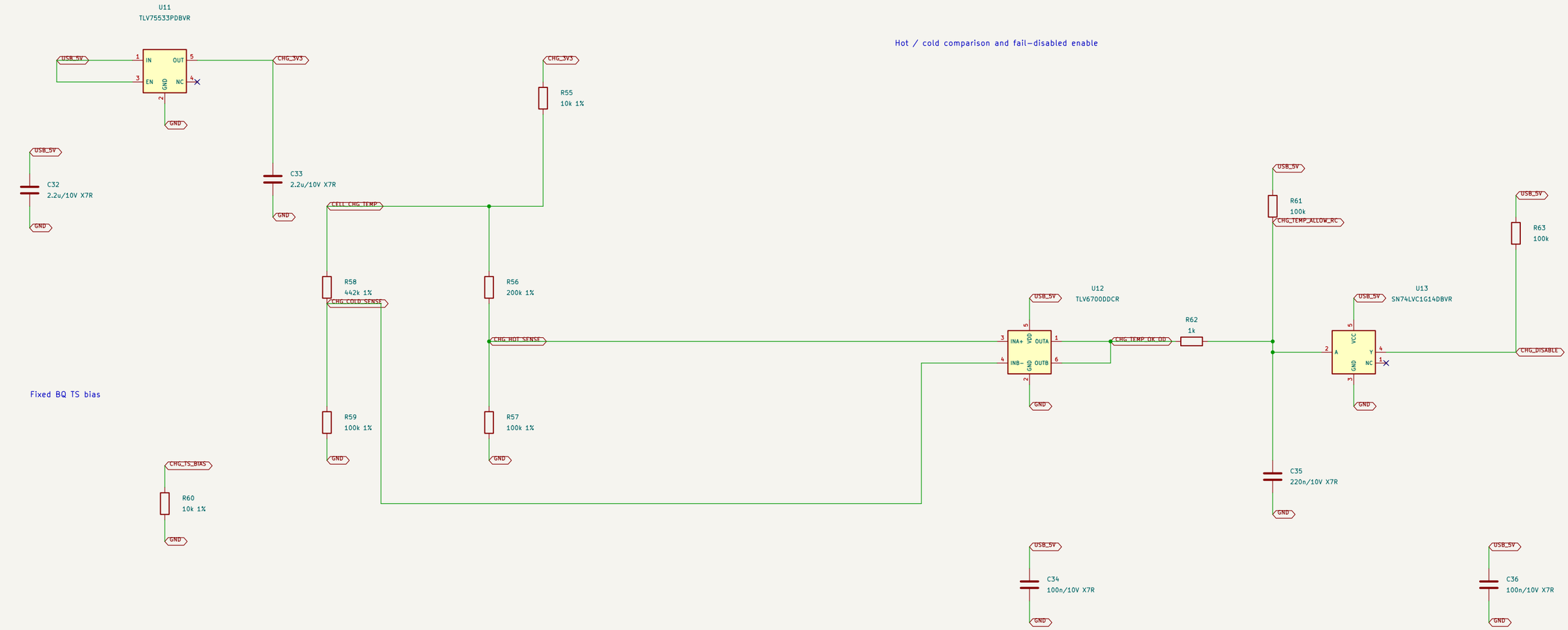
# Independent cell temperature charging window

USB-powered reference supply

TH1 at J4: actual cell temperature

Hot / cold comparison and fail-disabled enable

Fixed BQ TS bias




TH1 is an electrically insulated OFF-BOARD 103AT-2, bonded to the cell. Open/short probe disables charging.
U12 open-drain outputs both release only within the temperature window. U13 then drives CE low.
R61/C35 delay charge enable at USB startup; R62 limits capacitor discharge into U12. No ESP32 dependency.
R60 intentionally replaces the direct TS thermistor. This independent circuit now decides cell charge temperature.
Tolerance calculation uses the NTC beta model: verify actual resistance-temperature curve and trip limits on a fixture.

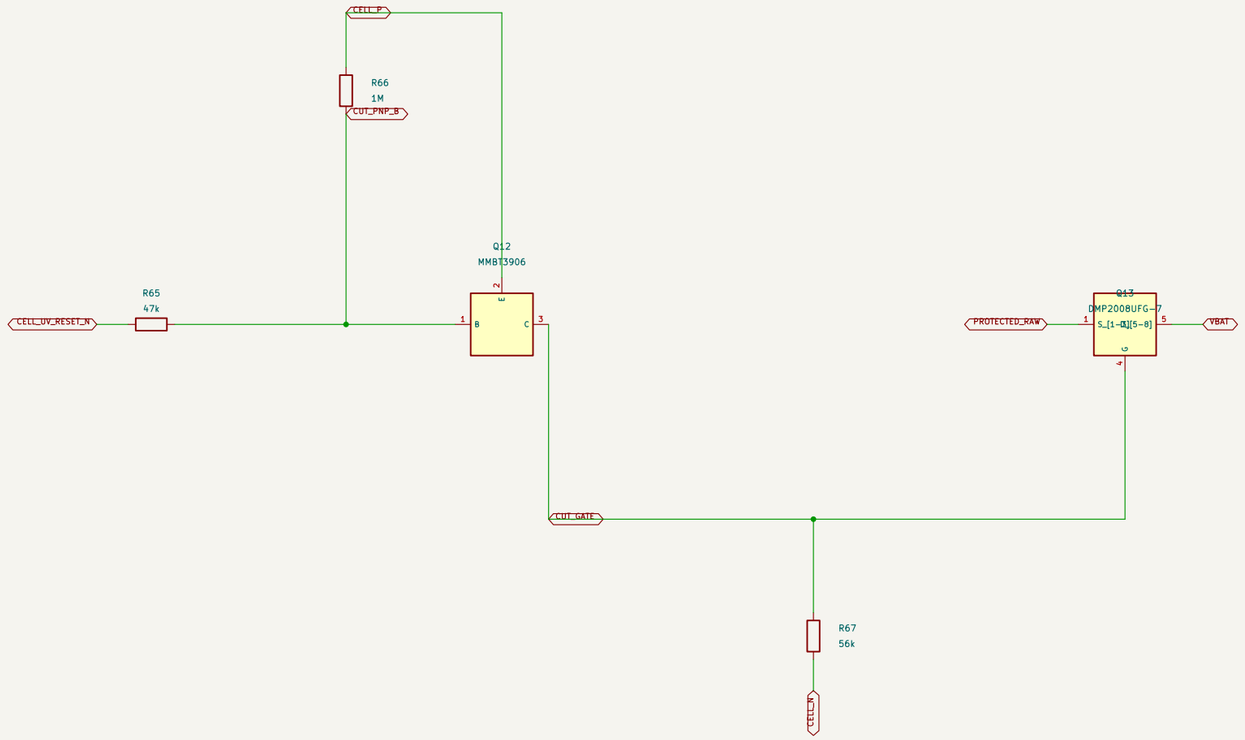
**Nate | Omni camera rover**

Sheet: /Cell temperature charging window/
File: charge_temperature.kicad_sch

***Title: Independent cell temperature charging window***

| Size: A3 | Date: | **Rev: E – prototype** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 10/11 |

# Discharge cutoff with charge recovery




Discharge: CELL_P -> Q7/Q8 -> PROTECTED_RAW -> Q13 -> VBAT

U10 RESET_N low turns Q12 on; Q13 gate rises to raw CELL_P and blocks discharge.
Static trip window 3.0535..3.1465 V; recovery approximately 3.3 V. Verify dip response on a cell simulator.
Q13 body diode points VBAT -> PROTECTED_RAW, permitting recovery charging while discharge is off.
Q13 MUST be downstream of Q7/Q8: lifting Q7 source could exceed its gate-source rating at startup.
U7 BAT/VDD still monitor raw cell; PACK and R41 connect to PROTECTED_RAW before Q13.
BQ24074 and independent cell-mounted TH1 window still control charging; U7 protection stays in series.
U7 cold-start/charger-wake compatibility remains a separate fabrication blocker. This is not a storage disconnect.

**Nate | Omni camera rover**

Sheet: /Independent discharge cutoff/
File: discharge_cutoff.kicad_sch

***Title: Discharge cutoff with charge recovery***

| Size: A3 | Date: | **Rev: E – prototype** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 11/11 |